

Tomáš Jíra, Ph.D.

✉ tomas.jira@vscht.cz

📍 Prague, Czech Republic

🆔 0009-0008-6143-1011

🌐 tjira

Education

Ph.D.	University of Chemistry and Technology, Prague , Chemistry	Prague, Czech Republic
	• Supervisor: Prof. Petr Slavíček • Dissertation: <i>Computational Strategies for Nonadiabatic Molecular Dynamics</i>	2022 – 2026
Ing.	University of Chemistry and Technology, Prague , Physical Chemistry	Prague, Czech Republic
		2020 – 2022
Bc.	University of Chemistry and Technology, Prague , Chemical Engineering	Prague, Czech Republic
		2017 – 2020

Experience

	University of Chemistry and Technology, Prague , Research/Teaching Position	Prague, Czech Republic
	• Research activities in the field of theoretical chemistry and photodynamics. • Teaching undergraduate mathematics courses.	2022 – present 4 years
	Christian-Albrechts-Universität zu Kiel , Research Internship (Kiel group)	Kiel, Germany
	• Group of Theoretical Spectroscopy and Heterogeneous Catalysis for solids, surfaces and condensed systems. • Supervisor: Prof. Anna Hehn	June 2026 – June 2026 1 month

Publications

Critical Assessment of Curvature-Driven Surface Hopping Algorithms	2025
T. Jíra, J. Janoš, P. Slavíček 10.1021/acs.jctc.5c01176 (Journal of Chemical Theory and Computation)	
Spin-orbit induced charge transfer in $\mathrm{H}^+/\mathrm{D}^+/\mathrm{T}^+ - \mathrm{N}(\mathrm{S})$ collisions	2025
M. Buchowiecki, T. Jíra 10.1016/j.jqsrt.2025.109697 (Journal of Quantitative Spectroscopy and Radiative Transfer)	
Sensitivity Analysis in Photodynamics: How Does the Electronic Structure Control <i>cis</i>-Stilbene Photodynamics?	2024
T. Jíra, J. Janoš, P. Slavíček 10.1021/acs.jctc.4c01008 (Journal of Chemical Theory and Computation)	

Projects

Zinq

Code for quantum dynamics, mixed quantum-classical dynamics, and electronic structure methods.